**UNIVERSITI TEKNIKAL MALAYSIA MELAKA**

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| **ASSIGNMENT 1**<br><br>**BERL3125 – SYSTEM INTEGRATION DESIGNING**<br>CLO 2: Construct and manage structure of an automation system, configuration and parameterization of its hardware and software (PLO 2, P5). |  |

| Name | Matrix No |
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| Week 9~14 Task: SCADA HMI of industrial process |
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| The objective of this assignment is to provide students with a practical opportunity to develop a SCADA HMI of an industrial process by using SCADA software. The HMI is a Level 2 or 3 HMI (based on ISA-101) which should include instrumentation devices as well as automated sequence of operations.<br><br>In this assignment:<br>• Create a new SCADA project or you can simulate any <u>one</u> SCADA project examples from Intouch Wonderware project simulations<br>• Develop the SCADA HMI project simulation with –<br>    o minimum of **3 (THREE)** process instruments with some motions/flows<br>    o instrumentation devices such as sensors also shown<br>    o indicators such as sliders, trend or alarm to are also included<br>    o window script coding are required<br>This is an individual assignment.<br>Upon completion of this assignment, the student should:<br>• demonstrate the HMI simulation screen to the lecturer on an agreed upon date<br>• Submit a report describing and displaying a few screens of your HMI operation sequences. |

## 1. Introduction

The objective of this project is to develop a **Level 2 HMI** (ISA-101) simulation using **Wonderware InTouch**. The project simulates an **industrial process** where a fluid is selected based on temperature readings. The HMI includes process instruments, sensors, indicators, and automated sequences to demonstrate control logic and system monitoring.

## 2. Project Objectives

Develop an HMI simulation with a **minimum of three process instruments**.

Simulate **temperature control and fluid selection** using a slider.

Demonstrate **instrumentation devices** like sensors and valves.

Include **indicators** such as sliders, trends, and alarms.

Implement **window script coding** for valve operation logic.

Showcase **automated sequence of operations** for fluid selection.

## 3. System Overview

The system simulates a **fluid filling station** based on temperature ranges:

**Temperature Slider:** Simulates the process temperature input.

**Temperature Sensor:** Reads the temperature from the slider.

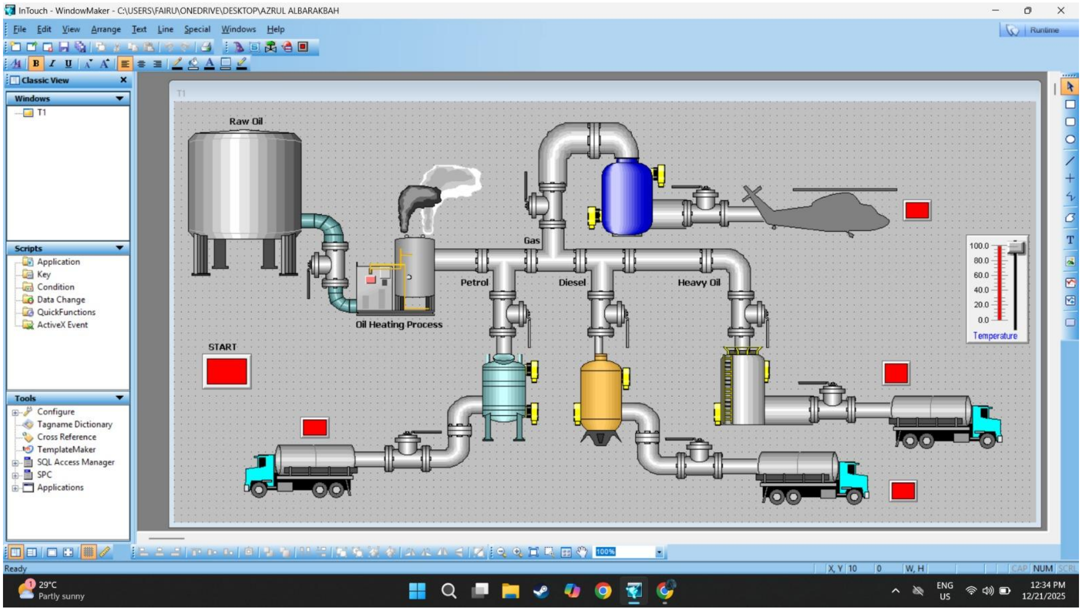
**Valves:** Four valves correspond to different fluids:

- Gas
- Petrol
- Diesel
- Heavy Oil

**Indicators:** Lamps for valve status, numeric display of temperature, and optional trends for temperature monitoring.

**Automation Logic:** Only one valve is open at a time, based on temperature.

## 4. HMI Design Screen




## 5. Process Instrumentation / Animation Screen

Start button is pressed for ON rawoilvalve to transfer raw oil to oil heating process.

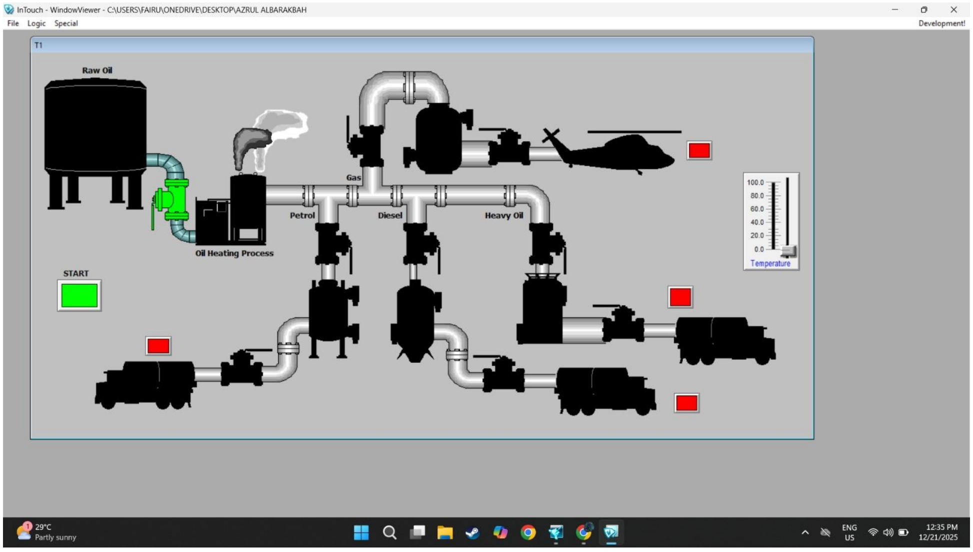



Figure1

Temperature controlled by slider. When temperature are between 10 and 25 valveGas ON for transfer gas from heating process to gas tank.

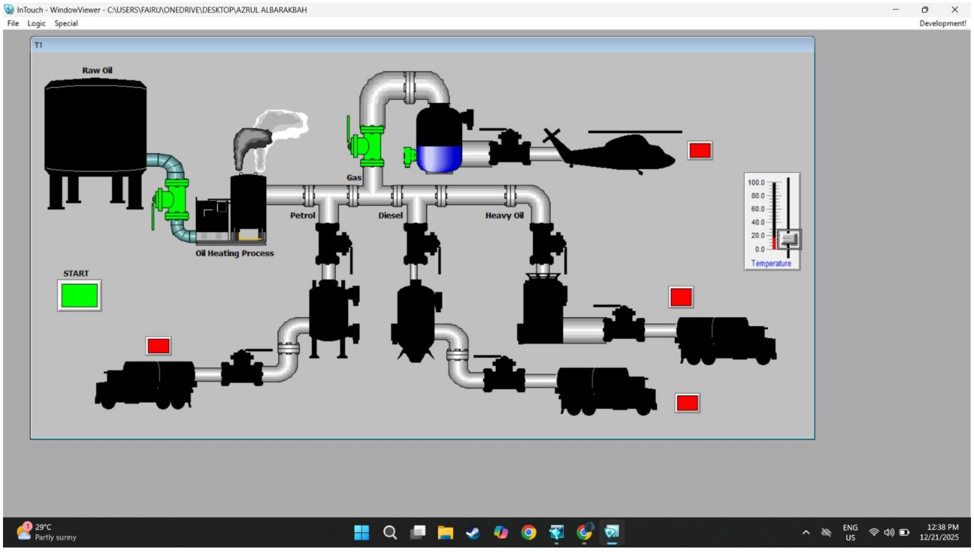



Figure2

When sensor1 HIGH valveGas OFF and valveGasTruck ON to fill the truck with gas in tank.

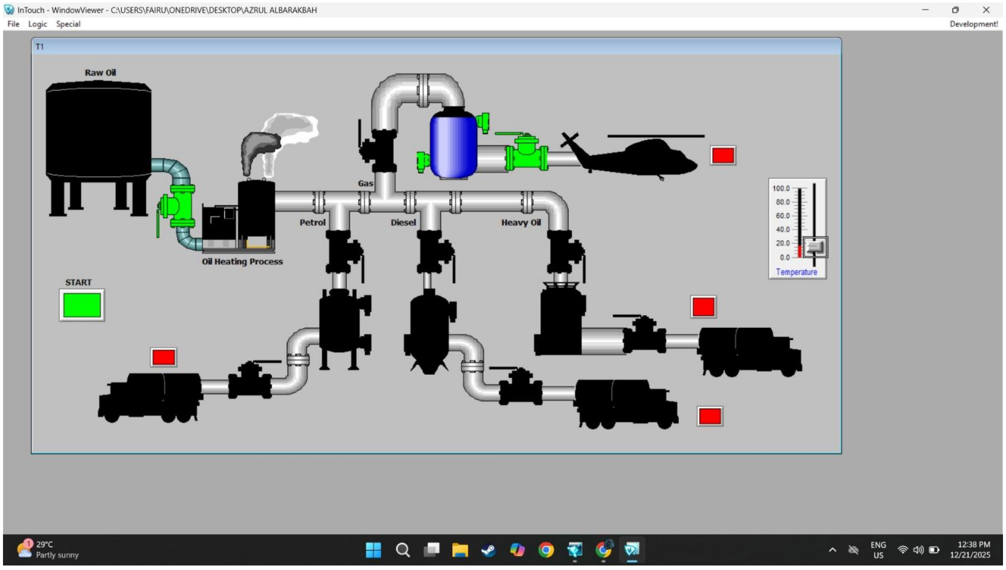



Figure3

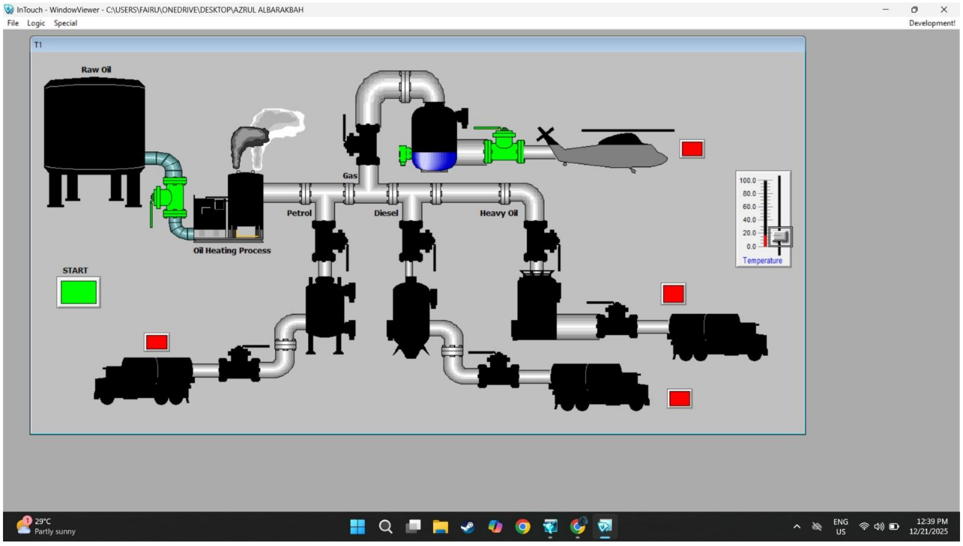



Figure4

When tank is full valveGasTruck OFF to stop filling truck until reset button pressed.

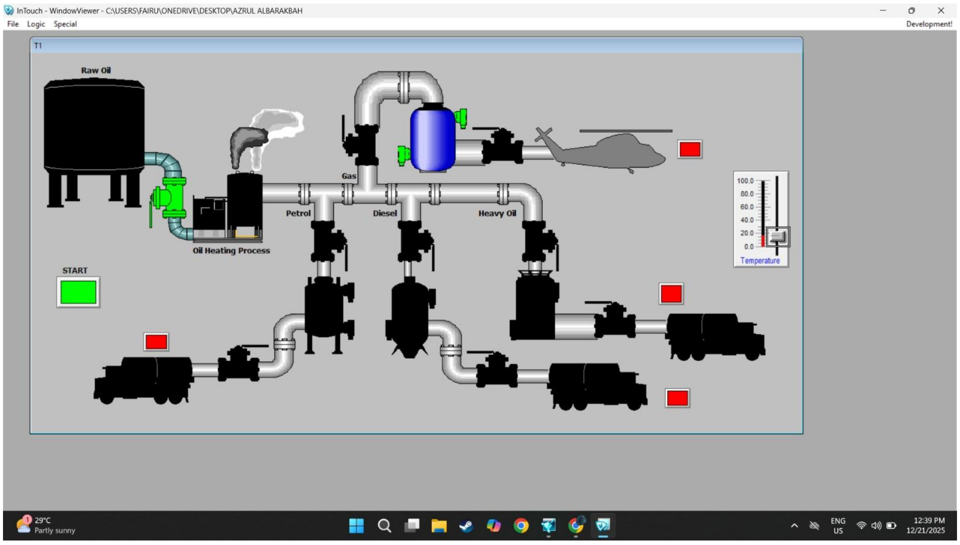



Figure5

When reset gas button pressed, gas in truck reset and gas in tank automatically transfer to truck again.

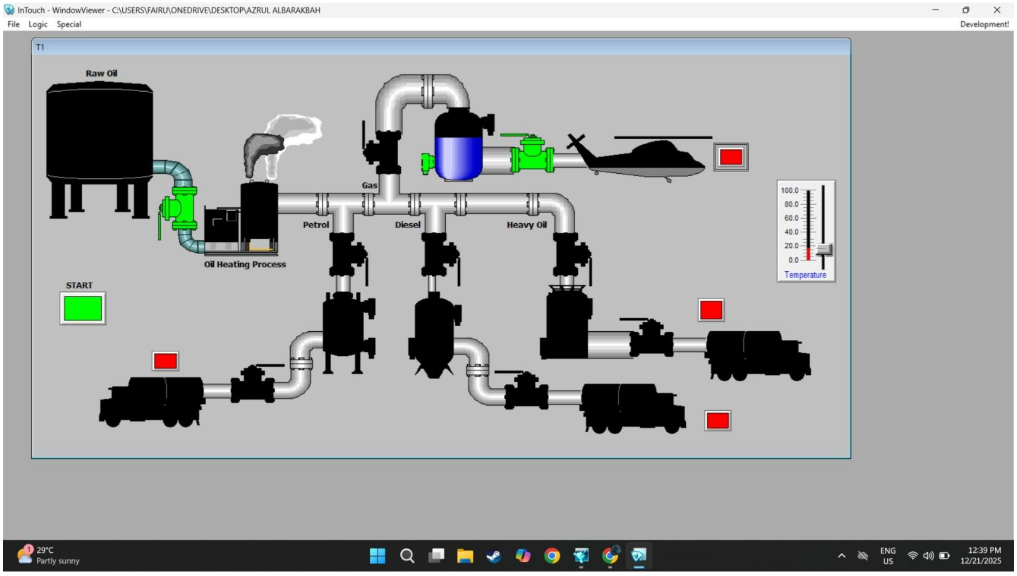



Figure6

All the process from gas are same to petrol, diesel and heavy oil process. Figure7 show if temperature are between 26 and 50 degree.

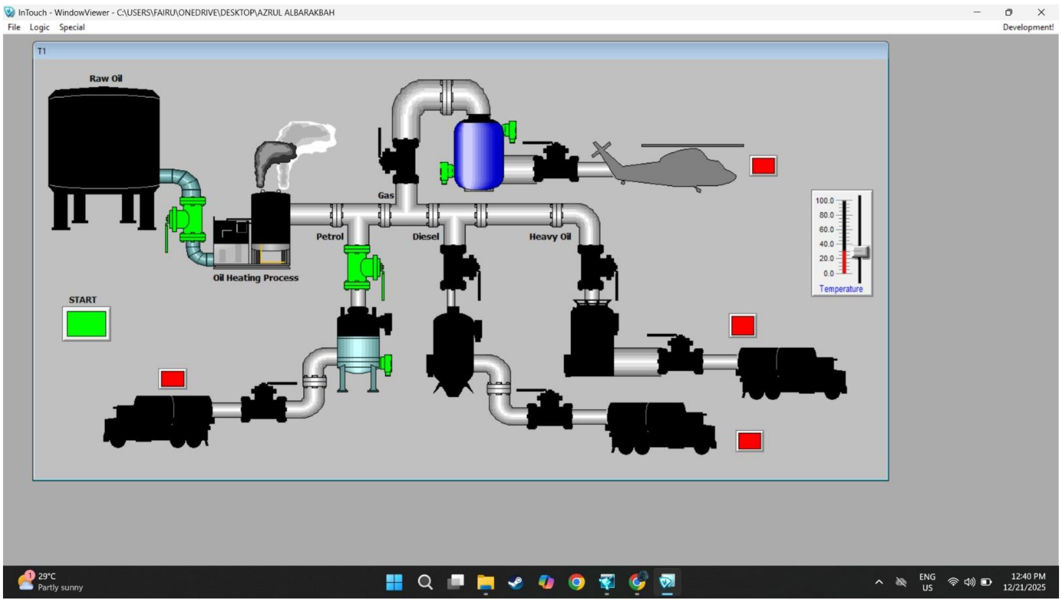



Figure7

Figure8 show if temperatured are between 51 and 75.

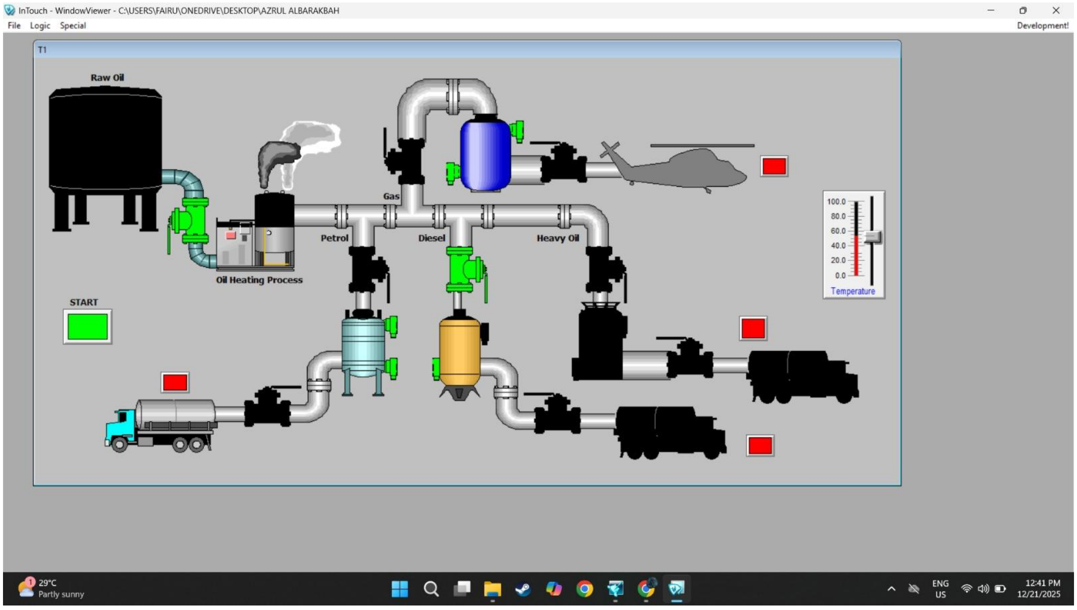



Figure8

Figure9 show if temperature are more than 75 degree.

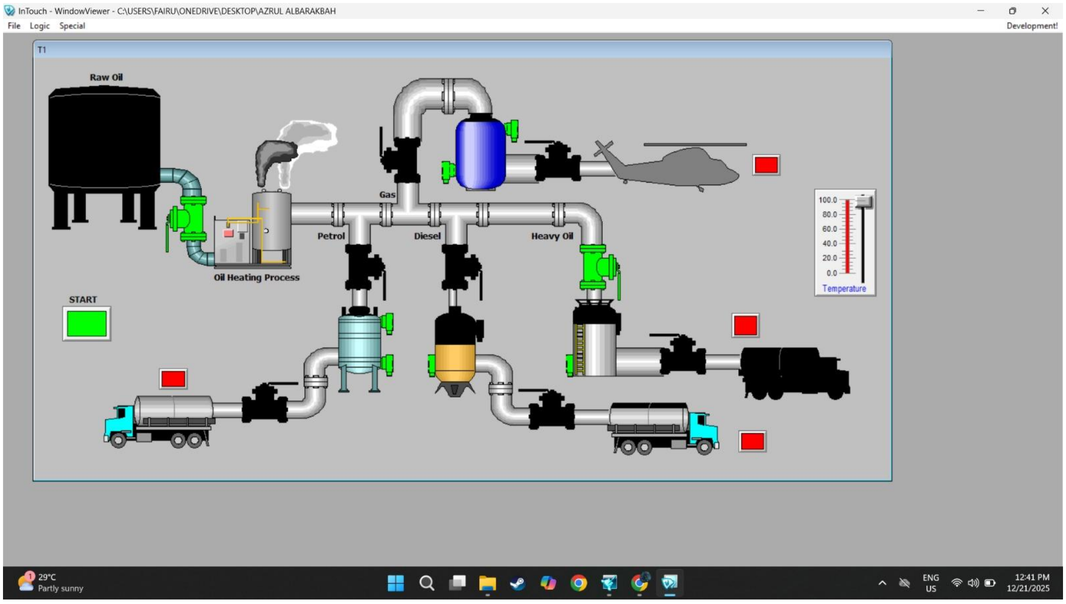



Figure9

Figure10 show the system when all truck and tank in full condition

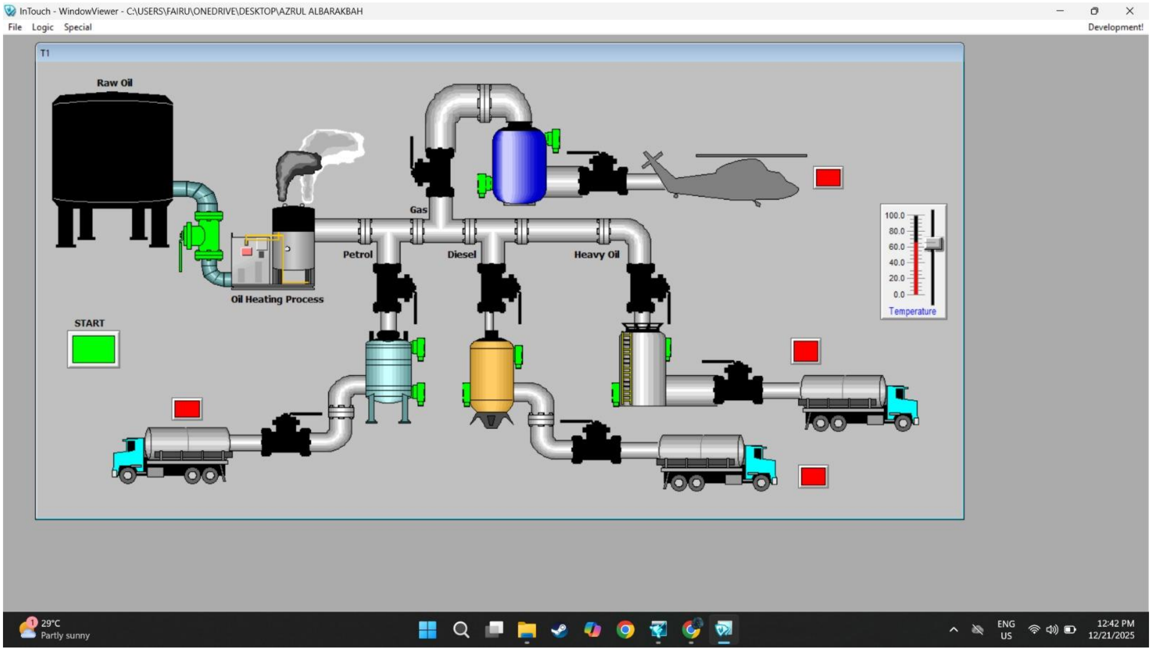



Figure10

## 6. Control Logic / Window Script

ON Startup

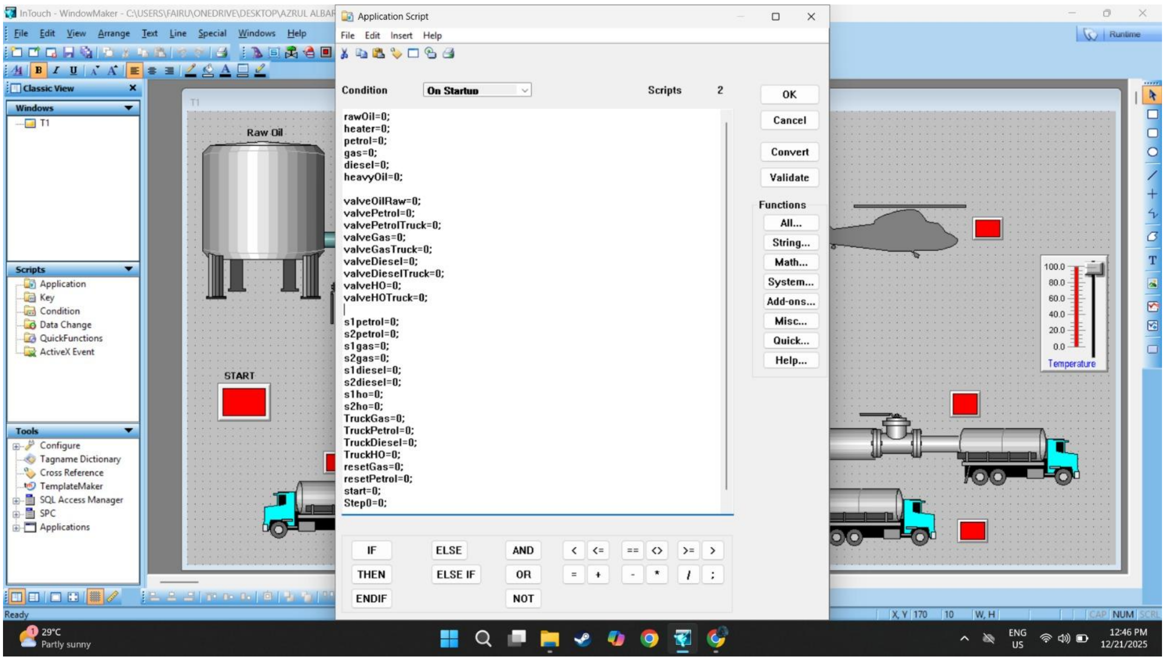



Figure11: ON Startup

While Running

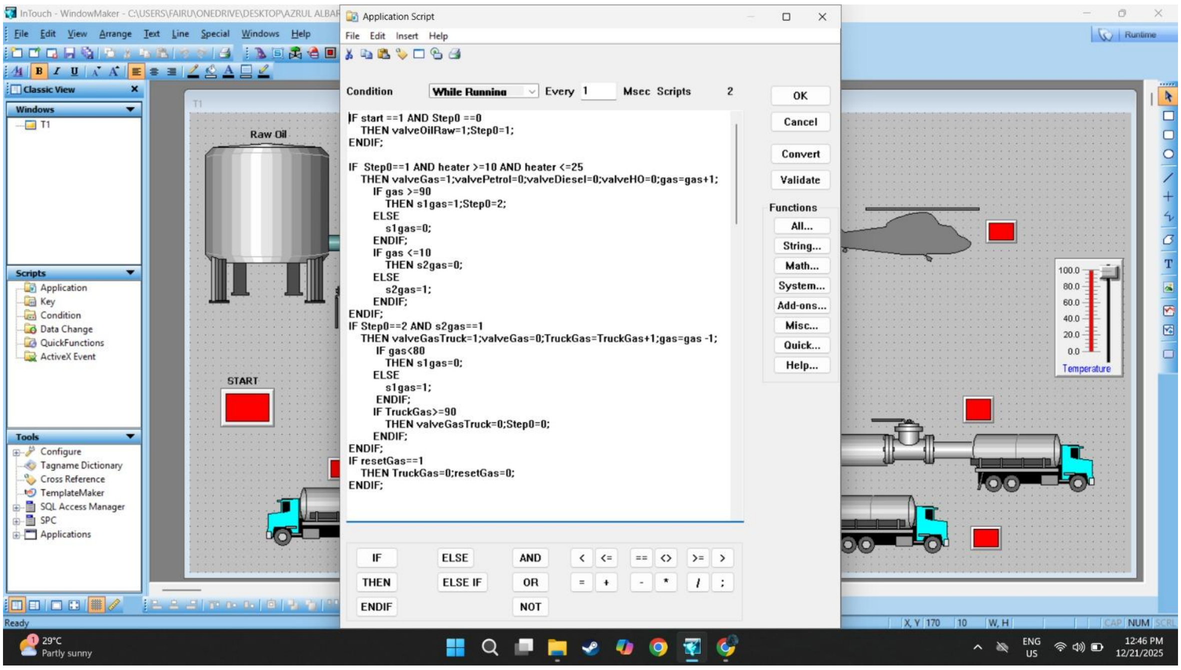



Figure12:Gas

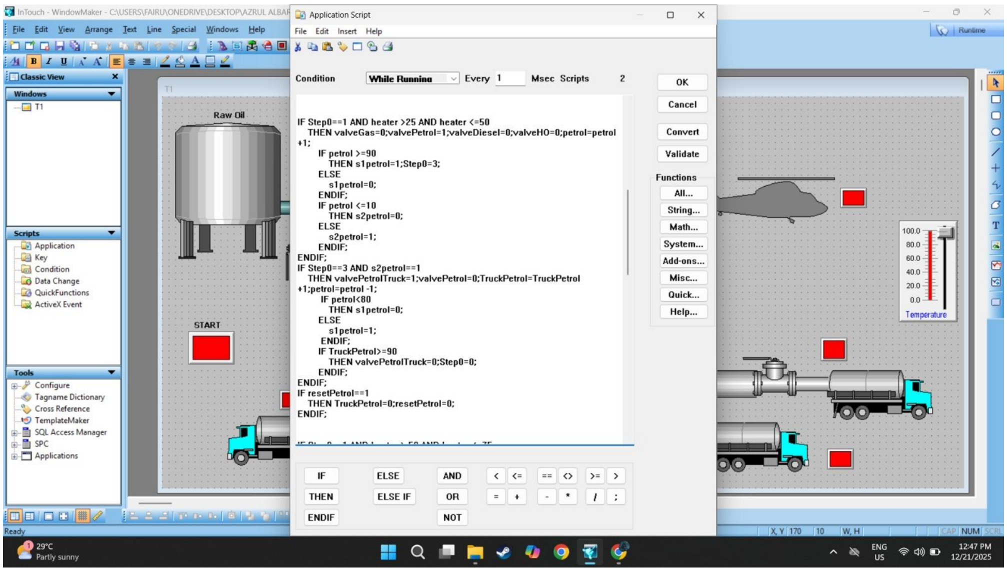



Figure13:Petrol

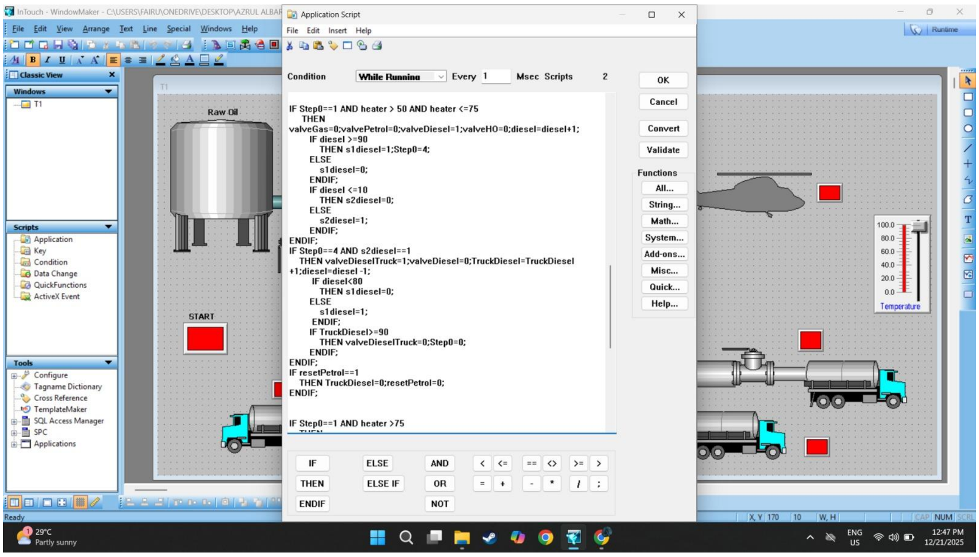



Figure14:Diesel

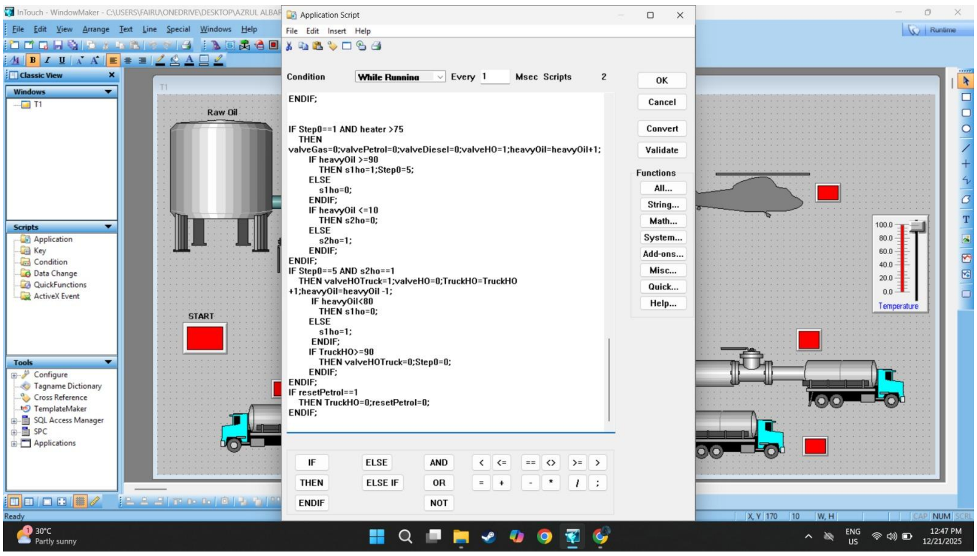



Figure15:Heavy Oil

Tag

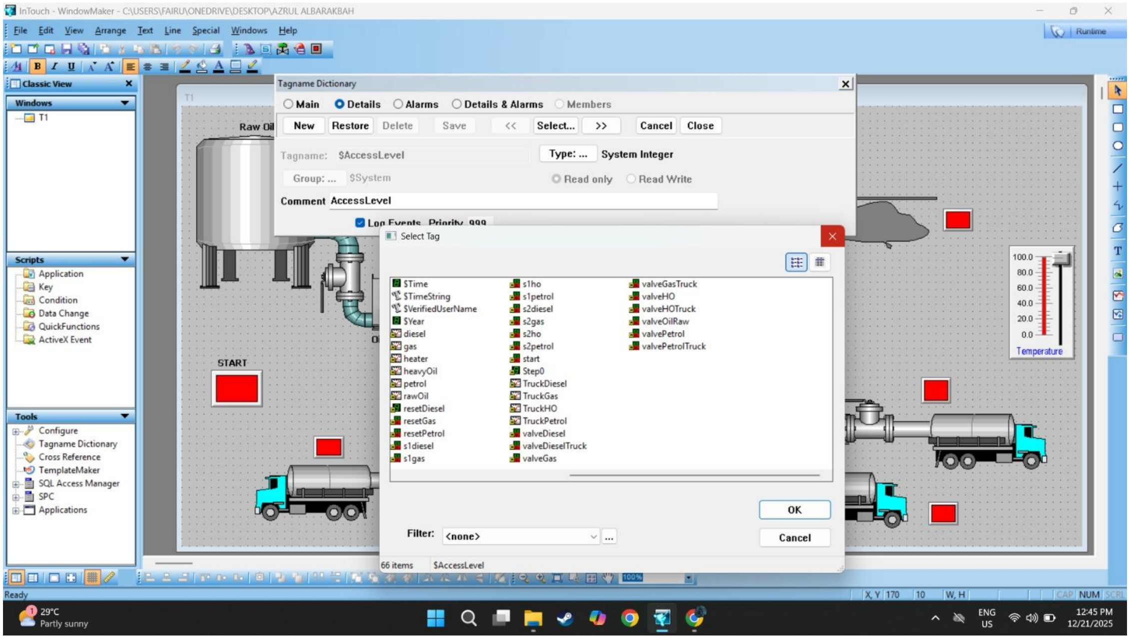



Figure16:Tag

## 7. Conclusion

In conclusion, this SCADA HMI project successfully demonstrates a temperature-based fluid selection system using a slider input, where the correct valve (Gas, Petrol, Diesel, or Heavy Oil) is automatically activated according to predefined temperature ranges. The HMI incorporates instrumentation devices, indicators, and automated sequences to provide clear visualization and control of the process, ensuring operational safety and accurate fluid selection. The project meets the assignment requirements and provides a practical simulation of industrial process monitoring and control.

**DETAILS FOR ASSIGNMENT ASSESSMENT RUBRICS**

| No | Items | Fully Met (4) | Met (3) | Partially Met (2) | Not Met (1) | Wght | Score | Mark |
|---|---|---|---|---|---|---|---|---|
| 1 | Project functionality | The project works and meets all of the specifications | The project displays and produces correct results and meets most specifications | The project produces partially correct results. | The project does not work or meet specifications. | 2.5 | | |
| 2 | Visibility / Complexity | The project is very well organized and at appropriate technical level | The project is fairly well organized or at fair technical level | The project is somewhat organized or at acceptable technical level | The project lacks organization or at low technical level | 2.5 | | |
| 3 | Usage of engineering tools | Excellent use of techniques and tools (proficiency) | Good use of techniques and tools | Able to use techniques and tools correctly | Poor use of techniques and tools (awareness) | 2 | | |
| 4 | Articulation | Articulate, logical, and interesting | Clear, logical and easy to follow | Understandable but not easy to follow | Cannot understand and no sequence | 2 | | |
| | Enthusiasm | Self-confidence, positive feeling about topic | A bit tense but at ease with topic | Uncomfortable, having trouble about topic | Tension, uninterested about topic | 20 | | |

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| 5 | Response to question | Show excellent knowledge with good elaboration | Show good knowledge with some elaboration | Show acceptable knowledge without elaboration | Show poor grasp without responding to question | 1 | | |
| | Contributions | Facilitates development and perform all assigned tasks | Contribute to development and perform most assigned tasks | Contribute ideas and perform some assigned tasks | Provide ideas sometimes and does some work | 10 | | |
| | Working with members | Collaborate with team effectively, always listen and encourage | Consistently support team, listen and help | Usually support team, sometimes listen and help | Occasionally support team, does not participate | 10 | | |

*Mark = Weight ✖ Score

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